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#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);

    int pid = fork();

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        execlp(cmd, cmd, NULL);
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child completed\n");
    }

    return 0;
}

```

```

yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd_year/OS/2520030525_Practical$ gedit q1.c

* (gedit:466): WARNING **: 08:46:13.989: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
* (gedit:466): WARNING **: 08:46:13.989: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd_year/OS/2520030525_Practical$ gcc q1.c -o q1
yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd_year/OS/2520030525_Practical$ ./q1
Enter Linux command: ls
Parent PID: 2887
Child PID: 2888
1 q1.c
Child completed

```

```

yeshwanth@LAPTOP-N3S9K224:/mnt/d/2nd_year/OS/2520030525_Practical$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                48 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:         0-15
Vendor ID:                   AuthenticAMD
Model name:                   AMD Ryzen 7 250 w/ Radeon 780M Graphics
CPU family:                   25
Model:                       117
Thread(s) per core:          2
Core(s) per socket:          8
Socket(s):                   1
Stepping:                     2
BogoMIPS:                     6587.68
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse
                              e sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology
                              tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse
                              4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_leg
                              acy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmc
                              all fsgsbase bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma c
                              lflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves avx512_bf16
                              clzero xsaveerptr arat npt nrtp_save tsc_scale vmcb_clean flushbyasid decodeassists pausef
                              ilter pfthreshold avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bita
                              lg avx512_vpopcntdq rdpid fsrm
Virtualization features:
Virtualization:               AMD-V
Hypervisor vendor:            Microsoft

```

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Virtualization type: full
Caches (sum of all):
  L1d: 256 MiB (8 instances)
  L1i: 256 MiB (8 instances)
  L2: 8 MiB (8 instances)
  L3: 16 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Not affected
  Retbleed: Not affected
  Spec rstack overflow: Vulnerable: Safe RET, no microcode
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSSB-eIBRS Not affected; BHI Not affected
  Srbds: Not affected
  Taa: Mitigation; Clear CPU buffers
  Tsx async abort: Not affected
  Vmscape: Not affected
yeshwanth@LAPTOP-N359K224:/mnt/d/2nd_year/OS/2520030525_Practical$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
yeshwanth@LAPTOP-N359K224:/mnt/d/2nd_year/OS/2520030525_Practical$ ps
```

```
yeshwanth@LAPTOP-N359K224:/mnt/d/2nd_year/OS/2520030525_Practical$ ps
PID TTY TIME CMD
314 pts/0 00:00:00 bash
5413 pts/0 00:00:00 ps
yeshwanth@LAPTOP-N359K224:/mnt/d/2nd_year/OS/2520030525_Practical$ top
top - 08:55:37 up 9 min, 1 user, load average: 0.20, 0.13, 0.05
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7587.7 total, 6117.5 free, 614.1 used, 1009.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 6973.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24552	15500	11604	S	0.0	0.2	0:00.93	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	58392	16004	15096	S	0.0	0.2	0:00.20	systemd-journal
67	systemd+	20	0	22500	10530	11960	S	0.0	0.2	0:00.11	systemd-resolve
89	root	20	0	35208	12400	9236	S	0.0	0.2	0:00.68	systemd-udev
141	root	20	0	2888	2012	1884	S	0.0	0.0	0:00.07	chronyd-starter
146	root	20	0	4472	3096	2852	S	0.0	0.0	0:00.01	cron
151	message+	20	0	8820	5408	4700	S	0.0	0.1	0:00.07	dbus-daemon
169	root	20	0	44092	29796	14652	S	0.0	0.4	0:00.23	networkd-dispat
176	root	20	0	18432	9440	8196	S	0.0	0.1	0:00.12	systemd-logind
189	root	20	0	1792300	15256	12616	S	0.0	0.2	0:00.14	wsl-pro-service
191	syslog	20	0	220548	5880	4580	S	0.0	0.1	0:00.08	rsyslogd
261	_chrony	20	0	24248	11700	7180	S	0.0	0.2	0:00.00	chronyd
272	_chrony	20	0	12092	2404	1760	S	0.0	0.0	0:00.01	chronyd
278	root	20	0	123056	32600	18028	S	0.0	0.4	0:00.20	unattended-upgr
286	root	20	0	5492	2812	2628	S	0.0	0.0	0:00.01	agetty
311	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
312	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.02	Relay(314)
314	yeshwan+	20	0	6268	5596	3924	S	0.0	0.1	0:00.11	bash
317	root	20	0	8648	5604	4760	S	0.0	0.1	0:00.01	login
380	yeshwan+	20	0	22684	13612	10980	S	0.0	0.2	0:00.23	systemd
382	yeshwan+	20	0	22912	4092	2088	S	0.0	0.1	0:00.00	(sd-pam)
409	yeshwan+	20	0	6136	5476	3928	S	0.0	0.1	0:00.04	bash
470	yeshwan+	20	0	8704	5276	4728	S	0.0	0.1	0:00.06	dbus-daemon
474	yeshwan+	20	0	380956	8104	7456	S	0.0	0.1	0:00.02	at-spi-bus-laun
481	yeshwan+	20	0	8500	5048	4676	S	0.0	0.1	0:00.01	dbus-daemon
482	yeshwan+	20	0	476852	17736	14852	S	0.0	0.2	0:00.09	xdg-desktop-por
487	yeshwan+	20	0	304804	6680	6180	S	0.0	0.1	0:00.02	xdg-permission-
492	yeshwan+	20	0	618996	7632	6948	S	0.0	0.1	0:00.02	xdg-document-po
498	root	20	0	2792	2296	2184	S	0.0	0.0	0:00.00	fusermount3

312	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.02	Relay(314)
314	yeshwan+	20	0	6268	5596	3924	S	0.0	0.1	0:00.11	bash
317	root	20	0	8648	5604	4760	S	0.0	0.1	0:00.01	login
380	yeshwan+	20	0	22684	13612	10980	S	0.0	0.2	0:00.23	systemd
382	yeshwan+	20	0	22912	4092	2088	S	0.0	0.1	0:00.00	(sd-pam)
409	yeshwan+	20	0	6136	5476	3928	S	0.0	0.1	0:00.04	bash
470	yeshwan+	20	0	8704	5276	4728	S	0.0	0.1	0:00.06	dbus-daemon
474	yeshwan+	20	0	380956	8104	7456	S	0.0	0.1	0:00.02	at-spi-bus-laun
481	yeshwan+	20	0	8500	5048	4676	S	0.0	0.1	0:00.01	dbus-daemon
482	yeshwan+	20	0	476852	17736	14852	S	0.0	0.2	0:00.09	xdg-desktop-por
487	yeshwan+	20	0	304804	6680	6180	S	0.0	0.1	0:00.02	xdg-permission-
492	yeshwan+	20	0	618996	7632	6948	S	0.0	0.1	0:00.02	xdg-document-po
498	root	20	0	2792	2296	2184	S	0.0	0.0	0:00.00	fusermount3
501	yeshwan+	20	0	481968	23820	18884	S	0.0	0.3	0:00.11	xdg-desktop-por
510	yeshwan+	20	0	168688	7792	7140	S	0.0	0.1	0:00.01	at-spi2-registr
2092	yeshwan+	20	0	165220	6008	6084	S	0.0	0.1	0:00.01	dconf-service
5423	yeshwan+	20	0	8288	6100	3936	R	0.0	0.1	0:00.07	top